

0	7	.	1
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Name the main carbohydrate transported through the phloem.

[1 mark]

0	7	.	2
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In the storage tissues of plants, carbohydrates transported through the phloem can be converted into starch.

Explain **three** features of starch that make it a good storage molecule.

[3 marks]

1

2

3

Question 7 continues on the next page

Turn over ►



Scientists investigated the mass flow hypothesis in geranium plants.

The scientists:

- separated the phloem from the xylem in a part of the stem by placing an impermeable layer between them
- enclosed a leaf above the separated section, and supplied it with radioactive carbon dioxide ($^{14}\text{CO}_2$)
- illuminated the plant for 15 hours
- cut the stem into sections as shown in **Figure 9**
- measured the radioactivity in the phloem and xylem of each section.

Figure 9

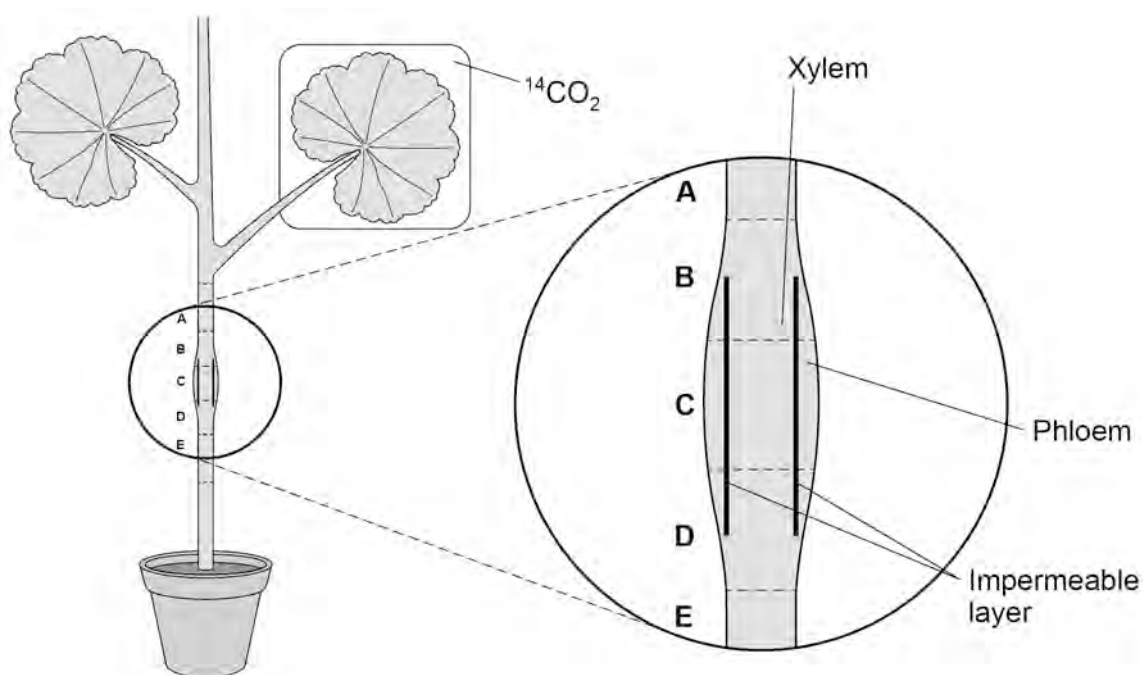


Table 4 shows the scientists' results.

Table 4

Section of tissue	Radioactivity in phloem / counts per minute	Radioactivity in xylem / counts per minute
A	41 100	24 210
B	2 761	755
C	130	74
D	101	47
E	157	75

